Haoying Zhang

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LinkedIn Profile | Github (Master projects) | GitLab (PhD projects)

Palaiseau, Ile de France - 91300, France

Education

- **INSA CVL and Inria de Saclay** *September 2023 - August 2026*
  PhD in Computer Science Palaiseau, France
  - Title: Privacy-Preserving Analysis and Sharing of Time Series Data
  - **Supervisors:** Benjamin Nguyen, Nicolas Anciaux

- **Institut National des Sciences Appliquées (INSA)** *September 2018 - June 2023*
  Engineering Master's Degree (Double Degree with Master of Research in Cybersecurity) Rennes, France
  - Major: Computer Science (Cloud and Cybersecurity)
  - Grade: 15.17/20 (Rank: 3/74)

Research and Teaching Experience

- **INSA CVL and Inria de Saclay [🌐]** *September 2023 - August 2026*
  PhD Palaiseau, France
  - Time series privacy-preserving computing
  - Expert on constraint programming, privacy regulations, machine learning and deep learning, differential privacy and cryptographic verification
  - 6 Publications at CCS, PETS, VLDB, ICDM and ACISP

- **Université du Québec à Montréal (UQAM) [🌐]** *March 2026 – August 2026*
  Visiting Researcher Montreal, Canada
  - Scaling a constraint computing baseline method by using deep learning method
  - Sequence-to-sequence regression for Matrix Profile inversion
  - Evaluating reconstruction and re-identification risks on five physiological time series datasets

- **ENSTA Paris [🌐]** *September 2023 - August 2026*
  Teaching Assistant Palaiseau, France
  - Taught Oracle Database to second-year engineering students
  - Prepared materials (VMs, servers) and supported students with exercises and individual problem-solving
  - Co-supervised a master students research internship at Inria

- **INSA CVL [🌐]** *February 2023 - August 2023*
  Research Intern Bourges, France
  - Research internship with topic "Privacy-Preserving Graph Processing"
  - Developed a framework called PUZZLE to compute PageRank on large distributed graph in a privacy-preserving way (with Differential Privacy guarantee)

Engineering Experience

- **NAGRA NexGuard [🌐]** *May 2022 - August 2022*
  Intern Rennes, France
  - Engineering internship with topic "High Performance Computing for Watermarking Algorithms"
  - Single Instruction Multiple Data (SIMD) architectures with vector processors on x86 and ARM
  - Integrated SIMD architectures to achieve high performance on multiple platforms

- **Huawei [🌐]** *July 2021 - September 2021*
  Intern Beijing, China
  - TCP/IP routing protocol automatic testing

PUBLICATIONS

C=CONFERENCE, S=SUBMISSION

[S.1] Haoying Zhang, et al. (2026). **Can Matrix Profiles Be Shared? Auditing Privacy-Utility Trade-offs in Physiological Time Series Analytics.**. Under review.

[C.6] Charlène Jojon, Haoying Zhang, et al. (2026). **Verifiable Anomaly and Similarity Detection Using Matrix Profile in Private Time-series**. Accepted for publication in Australasian Conference on Information Security and Privacy **(ACISP)** 2026.

[C.5] Haoying Zhang, et al. (2026). **Privacy Attacks on Matrix Profiles via Reconstruction Techniques** [Artifact: Available, Functional, Reproduced]. Accepted for publication in **PETS** 2026.

[C.4] Haoying Zhang, et al. (2025). **Exploring Utility and Attackability Trade-offs in Local Differential Privacy**. Accepted for publication in **CCS** 2025.

[C.3] Haoying Zhang, et al. (2025). **TELESAFE: Monitoring Energy Consumption for Work-Life Boundaries in Telework**. Accepted for publication in **ICDM** 2025.

[C.2] Haoying Zhang, et al. (2025). **TELESAFE: Detecting Private/Work Boundary Crossings in Energy Consumption Trails in Telework**. Accepted for publication in **VLDB** 2025.

[C.1] Haoying Zhang, et al. (2026). **Puzzle: Leveraging Synthetic Graph for Distributed Privacy-preserving Graph Processing**. Accepted for publication in **BDA** 2024.

SKILLS

- **Programming Languages:** Assembly, **C**, **C++**, **Python**, Java, **Rust**, OCaml, Perl, Ruby, JavaScript, PHP
- **Database Systems:** MySQL, Oracle
- **Machine Learning:** Classification algorithms, RNNs, CNNs, GANs, etc
- **Cloud & Containerization:** Docker, Kubernetes
- **DevOps:** Git
- **Privacy & Security:** Differential privacy, zero-knowledge proofs, privacy attacks (reconstruction, re-identification, membership inference), Federated Learning
- **Biomedical signals:** ECG, EEG processing and feature extraction

PROJECTS

- **Mycelium** *September 2021 - June 2022*
  Tools: Embedded system, C++, LoRaWAN, Raspberry Pi, Docker, Kubernetes [🌐]
  - Implemented an alarm system in sensors to trigger actions based on weather scenarios
  - Developed a Kubernetes-based fog infrastructure on a Raspberry Pi cluster for storing and processing sensor data transmitted via LoRaWAN, enabling monitoring of natural environments in Rennes
  - Collaborated with scientists and engineers from Geosciences Rennes and Clermont-Ferrand University
- **Exchange Rate Reminder** *August 2020*
  Tools: Python, InfluxDB [GitHub]
  - Implemented an email notification system triggered by real-time exchange rate anomalies
- **Chessboard game** *August 2020*
  Tools: JavaFx [GitHub]

VOLUNTEER EXPERIENCE

- **Mathematics and Physics Tutor** *September 2018 - February 2019*
  INSA
  - Tutored fellow students in mathematics and physics through classes and Q&A sessions
- **Volunteer at the Planetarium** *August 2014*
  Beijing Planetarium
  - Guided visitors through exhibits and provided telescope demonstrations

ADDITIONAL INFORMATION

Languages: Chinese (Mother tongue), French (Fluent: DELF B2, 85/100), English (Fluent: TOEIC, 945/990)
Interests: Snowboarding, Badminton, Running, Astronomy